



## Tutorial - 3

### Fourier Series.

Q.1 Find Fourier series of  $f(x) = x^2$  in  $(0, 2\pi)$  hence deduce that  $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$

$$\Rightarrow a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} x^2 dx = \frac{1}{2\pi} \left[ \frac{x^3}{3} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} \left[ \frac{8\pi^3}{3} - 0 \right] = \frac{4\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \cos nx dx =$$

$$\therefore \begin{array}{ll} u = x^2 & v = \cos nx \\ u' = 2x & v_1 = \sin nx / n \\ u'' = 2 & v_2 = -\cos nx / n^2 \\ u''' = 0 & v_3 = -\sin nx / n^3 \end{array}$$

$$\int u \cdot v dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

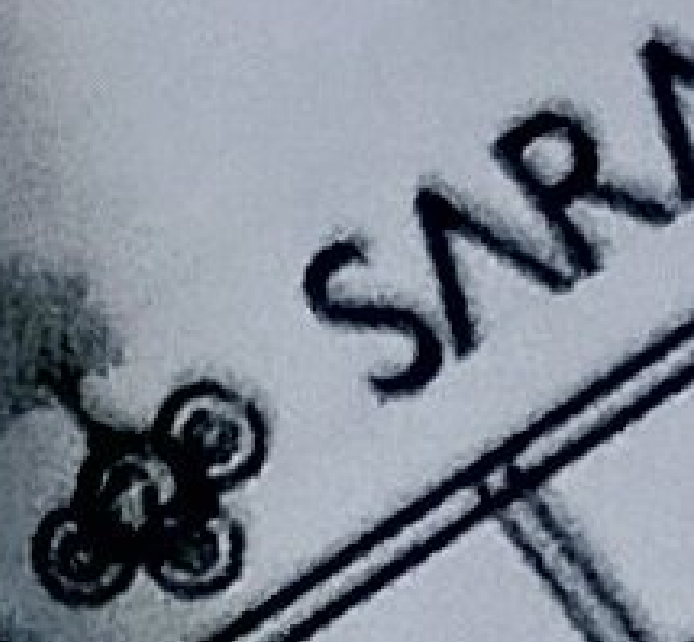
$$\therefore a_n = \frac{1}{\pi} \left[ \frac{x^2 \sin nx}{n} + \frac{2x \cos nx}{n^2} - \frac{2 \sin nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[ (0 + \frac{4\pi}{n^2} - 0) - (0) \right]$$

$$= \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx dx$$



$$\therefore u = x^2$$

$$v = \sin nx$$

$$u' = 2x$$

$$v_1 = -\cos nx / n$$

$$u'' = 2$$

$$v_2 = -\sin nx / n^2$$

$$u''' = 0$$

$$v_3 = \cos nx / n^3$$

$$\int u \cdot v \, dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \dots$$

$$\therefore b_n = \frac{1}{\pi} \left[ x^2 \left( \frac{-\cos nx}{n} \right) + 2x \frac{\sin nx}{n^2} + 2 \frac{\cos nx}{n^3} \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \left[ \left( \frac{-4\pi^2}{n} + 0 + \frac{2}{n^3} \right) - \left( \frac{2}{n^3} \right) \right]$$

$$= \frac{-4\pi}{n}$$

Fourier series for  $(0, 2\pi)$  is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$= \frac{4\pi^2}{3} + \sum \left( \frac{4}{n^2} \right) \cos nx + \sum \left( \frac{-4\pi}{n} \right) \sin nx$$

Put  $x = \pi$

$$\pi^2 = \frac{4\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} (-1)^n + 0$$

$$\frac{\pi^2 - 4\pi^2}{3} = 4 \left[ \frac{-1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots \right]$$

$$\frac{-\pi^2}{3} = -4 \left[ \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right]$$

$$\frac{\pi^2}{12} = \left[ \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots \right]$$

Q.2 Find Fourier series of function  $f(x) = e^{-x}$ , where  $0 \leq x \leq 2\pi$

$$\Rightarrow \text{Let } f(x) = e^{-x} = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \dots 1)$$

## SARASWATI College of Engineering

$$\therefore a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} e^{-x} dx = \frac{1}{2\pi} \left[ -e^{-x} \right]_0^{2\pi}$$

$$= \frac{1}{2\pi} (-e^{-2\pi} + 1) = \frac{1 - e^{-2\pi}}{2\pi}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cos nx dx$$

$$\therefore \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$a_n = \frac{1}{\pi} \left[ \frac{e^{-x}}{1+n^2} \left[ \cancel{e^x} \cos nx (-\cos nx + n \sin nx) \right] \right]_0^{2\pi}$$

$$= \frac{1}{\pi} \frac{1}{1+n^2} \left[ e^{-x} (-\cos nx + n \sin nx) \right]_0^{2\pi}$$

$$= \frac{1}{\pi(1+n^2)} \left[ e^{-2\pi} (-\cos 2n\pi + n \sin 2n\pi) - e^0 (-\cos 0 + n \sin 0) \right]$$

$$= \frac{1}{\pi(1+n^2)} \left[ e^{-2\pi} (-1) - (-1) \right]$$

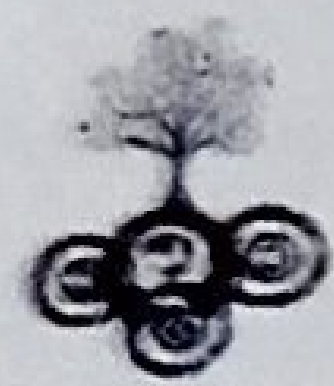
$$= \frac{1 - e^{-2\pi}}{\pi(1+n^2)}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin nx dx$$

$$\therefore \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$= \frac{1}{\pi} \frac{1}{1+n^2} \left[ e^{-x} (-\sin nx - n \cos nx) \right]_0^{2\pi}$$



$$= \frac{1}{\pi(1+n^2)} \left[ e^{-2\pi} (-\sin 2n\pi - n \cos 2n\pi) - e^0 (-\sin 0 - n \cos 0) \right]$$

$$= \frac{1}{\pi(1+n^2)} \left[ e^{-2\pi} (-n) - 1(-n) \right] = \frac{n}{\pi(1+n^2)} (1 - e^{-2\pi})$$

Putting the values in i)

$$e^{-x} = \frac{1 - e^{-2\pi}}{2\pi} + \frac{(1 - e^{-2\pi})}{\pi} \left[ \sum_{n=1}^{\infty} \frac{1}{1+n^2} \cos nx + \sum_{n=1}^{\infty} \frac{n}{1+n^2} \sin nx \right]$$

$$= \frac{1 - e^{-2\pi}}{\pi} \left[ \frac{1}{2} + \sum_{n=1}^{\infty} \frac{1}{1+n^2} (\cos nx + n \sin nx) \right]$$

Q.3 Find fourier series for  $f(x) = \frac{3x^2 - 6x\pi + 2\pi^2}{12}$  in  $(0, 2\pi)$  hence deduce that  $\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$

$\Rightarrow$  Let  $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$

$$\therefore a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \frac{(3x^2 - 6x\pi + 2\pi^2)}{12} dx$$

$$= \frac{1}{24\pi} \left[ \frac{3x^3}{3} - \frac{6x^2\pi}{2} + 2\pi^2 x \right]_0^{2\pi}$$

$$= \frac{1}{24\pi} [8\pi^3 - 12\pi^3 + 4\pi^3] = 0$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \frac{(3x^2 - 6x\pi + 2\pi^2)}{12} \cos nx dx$$

$$u = 3x^2 - 6x\pi + 2\pi^2/12$$

$$v = \cos nx$$

$$u' = x^2 - 6\pi$$

$$v_1 = \sin nx / n$$

$$u'' = 2x$$

$$v_2 = -\cos nx / n^2$$

$$u''' = 2$$

$$v_3 = -\sin nx / n^3$$

$$u'''' = 0$$

$$v_4 = \cos nx / n^4$$

$$a_n = \frac{1}{12\pi} \left[ 3 \int_0^{2\pi} x^2 \left( \frac{\sin nx}{n} \right) - (2x) \left( \frac{-\cos nx}{n^2} \right) + (2) \left( \frac{-\sin nx}{n^3} \right) \right. \\ \left. - 6\pi \int_0^{2\pi} x \left( \frac{\sin nx}{n} \right) - (1) \left( \frac{-\cos nx}{n^2} \right) \right. \\ \left. + 2\pi^2 \left( \frac{\sin nx}{n} \right) \right]_{0}^{2\pi}$$

$$= \frac{1}{12\pi} \left[ 3 \left\{ 0 + \frac{4\pi}{n^2} + 0 \right\} - 6\pi \left\{ 0 + \frac{1}{n^2} \right\} + 2\pi^2(0) - \right. \\ \left. \left\{ 0 - 6\pi \left( \frac{1}{n^2} \right) \right\} \right]$$

$$= \frac{1}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_0^{2\pi} \left( \frac{3x^2 - 6x\pi + 2\pi^2}{12} \right) \sin nx \, dx$$

$$= \frac{1}{12\pi} \left[ 3 \int_0^{2\pi} x^2 \left( \frac{-\cos nx}{n} \right) - (2x) \left( \frac{-\sin nx}{n^2} \right) + 2(1) \left( \frac{\cos nx}{n^3} \right) \right. \\ \left. - 6\pi \int_0^{2\pi} x \left( \frac{-\cos nx}{n} \right) - (1) \left( \frac{-\sin nx}{n^2} \right) \right. \\ \left. + 2\pi^2 \left( \frac{-\cos nx}{n} \right) \right]_{0}^{2\pi}$$



$$b_n = \frac{1}{12\pi} \left[ 3 \left\{ -\frac{4\pi^2}{n} + \frac{2}{n^3} \right\} - 6\pi \left\{ -\frac{2\pi}{n} \right\} - \frac{2\pi^2}{n} \right. \\ \left. - \left\{ 3 \left( \frac{2}{n^3} \right) - \frac{2\pi^2}{n} \right\} \right] = 0$$

$$\therefore f(x) = \sum a_n \cos nx = \sum \frac{1}{n^2} \cos nx$$

$$\therefore \frac{3x^2 - 6x\pi + 2\pi^2}{12} = \frac{1}{1^2} \cos x + \frac{1}{2^2} \cos 2x + \frac{1}{3^2} \cos 3x + \dots$$

Putting  $x=0$

$$\frac{2\pi^2}{12} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

$$\therefore \frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots$$

Hence proved.

Q.4 Find fourier series of  $f(x) = x^2$  in  $(-\pi, \pi)$  hence prove that

$$i) \quad \frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$iii) \quad \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

$$ii) \quad \frac{\pi^2}{12} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$

$$\Rightarrow f(-x) = (-x)^2 = x^2 = f(x)$$

Here  $f(x)$  is an even function

$$\therefore b_n = 0$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x^2 dx$$

$$= \frac{1}{\pi} \left[ \frac{x^3}{3} \right]_0^{\pi} = \frac{\pi^2}{3}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx \, dx$$

$$= \frac{2}{\pi} \left[ x^2 \left( \frac{\sin nx}{n} \right) - (2x) \left( \frac{-\cos nx}{n^2} \right) + 2 \left( \frac{-\sin nx}{n^3} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[ \left\{ 0 - (2\pi) \left( \frac{-\cos n\pi}{n^2} \right) + 0 \right\} - \{0\} \right]$$

$$= \frac{4}{n^2} \cos n\pi = \frac{4}{n^2} (-1)^n$$

Putting these values in 1)

$$x^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2} \quad \dots 2)$$

i) Put  $x = \pi$  in 2)

$$\therefore \pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n \cos n\pi}{n^2}$$

$$\pi - \frac{\pi^2}{3} = 4 \sum_{n=1}^{\infty} \frac{(-1)^n (-1)^n}{n^2} = 4 \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n^2}$$

$$\therefore \frac{2\pi^2}{3} \cdot \frac{1}{4} = \sum_{n=1}^{\infty} \frac{(-1)^{2n}}{n^2}$$

$$\therefore \frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

ii) Put  $x = 0$  in 1)

$$0 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n \cdot 1}{n^2}$$

$$\therefore -\frac{\pi^2}{3} = 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$$

$$\frac{\pi^2}{12} = (-1) \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$



Adding i) &amp; ii)

$$\frac{\pi^2}{6} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \dots$$

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \dots$$

$$\therefore \frac{\pi^2}{6} + \frac{\pi^2}{12} = 2 \left( \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right)$$

$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Q.5

Find fourier expansion of  $f(x) = x$  in  $-\pi \leq x \leq \pi$  $\Rightarrow$ 

$$f(-x) = -x = -f(x)$$

 $f(x)$  is odd function.

$$a_0 = 0 \quad \& \quad a_n = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \sin nx \, dx$$

$$u = x$$

$$v = \sin nx$$

$$u' = 1$$

$$v_1 = \frac{-\cos nx}{n}$$

$$v_2 = \frac{-\sin nx}{n^2}$$

$$u'' = 0$$

$$b_n = \frac{2}{\pi} \left[ x \cdot \left( \frac{-\cos nx}{n} \right) + \frac{\sin nx}{n^2} \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[ \frac{-\pi \cos n\pi}{n} + \frac{\sin n\pi}{n^2} - (0 + 0) \right]$$

$$= \frac{2}{\pi} \frac{-\pi(-1)^n}{n}$$

$$= \frac{2(-1)^n \pi}{\pi n} = \frac{2(-1)^{n+1}}{n}$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx$$

Q. 6

Find fourier expansion of  $f(x) = x^3$  in  $(-\pi, \pi)$  $\Rightarrow$ 

$$f(x) = x^3$$

$$f(-x) = (-x)^3 = -x^3 = -f(x)$$

$f(x)$  is odd function in  $(-\pi, \pi)$

$$a_0 = 0 \quad \& \quad a_n = 0$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x^3 \sin nx \, dx$$

$$u = x^3$$

$$v = \sin nx$$

$$u' = 3x^2$$

$$v_1 = -\cos nx / n$$

$$u'' = 6x$$

$$v_2 = -\sin nx$$

$$u''' = 6$$

$$n^2 \quad v_3 = \cos(nx)$$

$$u^{(4)} = 0$$

$$v_4 = \frac{\sin nx}{n^3}$$

$$b_n = \frac{2}{\pi} \left[ \frac{-\pi^3 (-1)^n}{n} \right]$$

$$= \frac{2}{\pi} \left[ \frac{-x^3 \cos nx}{n} + \frac{3x^2 \sin nx}{n^2} + \frac{6x \cos nx}{n^3} - \frac{6 \sin nx}{n^4} \right]$$

$$= \frac{2}{\pi} \left[ \frac{-\pi^3 (-1)^n}{n} + \frac{6\pi (-1)^n}{n^3} \right]$$

$$= \frac{2}{\pi} \left[ (-1)^n \left( \frac{-\pi^3}{n} + \frac{6\pi}{n^3} \right) \right]$$

$$= \frac{2\pi^2 (-1)^{n+1}}{n} + \frac{12(-1)^n}{n^3}$$

$$= (-1)^n \left( \frac{12}{n^3} - \frac{2\pi^2}{n} \right)$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx = \sum_{n=1}^{\infty} (-1)^n \left[ \frac{12}{n^3} - \frac{2\pi^2}{n} \right] \sin nx$$



Q.7

Find fourier series for  $f(x) = |x|$ ,  $-\pi \leq x \leq \pi$  $\Rightarrow$ 

$$f(x) = |x|$$

$$f(-x) = (-|x|) = |x| = f(x)$$

 $f(x)$  is even function in  $(-\pi, \pi)$ 

$$b_n = 0, \text{ as } f(x) = |x|$$

$$f(x) = -x$$

$$-\pi \leq x \leq 0$$

$$= x$$

$$0 \leq x \leq \pi$$

$$a_0 = \frac{1}{\pi} \int_0^{\pi} f(x) dx = \frac{1}{\pi} \int_0^{\pi} x dx$$

$$= \frac{1}{\pi} \left[ \frac{x^2}{2} \right]_0^{\pi} = \frac{1}{\pi} \frac{\pi^2}{2}$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} x \cdot \cos nx dx$$

$$= \frac{2}{\pi} \left[ x \left( \frac{\sin nx}{n} \right) - 1 \left( \frac{-\cos nx}{n^2} \right) \right]_0^{\pi}$$

$$= \frac{2}{\pi} \left[ \left( 0 + \frac{(-1)^n}{n^2} \right) - \left( 0 + \frac{1}{n^2} \right) \right]$$

$$= \frac{2}{\pi n^2} [(-1)^n - 1]$$

$$f(x) = \frac{2}{\pi} \int_0^{\pi} f \quad a_0 + \sum a_n \cos nx$$

$$= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} \cos nx$$

Q.8 Find fourier expansion of  $f(x) = 4 - x^2$  in  $(0, 2)$ .  
 $\Rightarrow$  Here  $2l = 2 \quad \therefore l = 1$

Let  $f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$

$$= a_0 + \sum_{n=1}^{\infty} a_n \cos n\pi x + \sum_{n=1}^{\infty} b_n \sin n\pi x$$

$$a_0 = \frac{1}{2l} \int_0^{2l} f(x) dx = \frac{1}{2} \int_0^2 (4 - x^2) dx$$

$$= \frac{1}{2} \left[ 4x - \frac{x^3}{3} \right]_0^2 = \frac{1}{2} \left[ 8 - \frac{8}{3} \right]$$

$$= \frac{8}{3}$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos \frac{n\pi x}{l} dx$$

$$= \int_0^2 (4 - x^2) \cos n\pi x dx$$

$$\therefore u = 4 - x^2$$

$$v = \cos n\pi x$$

$$u' = -2x$$

$$v_1 = \sin n\pi x / n\pi$$

$$u'' = -2$$

$$v_2 = -\cos n\pi x / n^2 \pi^2$$

$$u''' = 0$$

$$v_3 = -\sin n\pi x / n^3 \pi^3$$

$$\int u \cdot v dx = \int u v_1 - u' v_2 + u'' v_3 - u''' v_4 \dots$$

$$a_n = \left[ (4 - x^2) \left( \frac{\sin n\pi x}{n\pi} \right) - (-2x) \left( \frac{-\cos n\pi x}{n^2 \pi^2} \right) + (-2) \left( \frac{-\sin n\pi x}{n^3 \pi^3} \right) \right]_0^2$$

$$= \left[ \left\{ 0 - \frac{4}{n^2 \pi^2} + 0 \right\} - \left\{ 0 \right\} \right] = -\frac{4}{n^2 \pi^2}$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin \frac{n\pi x}{l} dx = \int_0^2 (4 - x^2) \sin n\pi x dx$$

$$u = 4 - x^2$$

$$v = \sin n\pi x$$

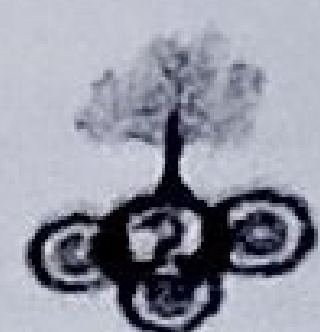
$$u' = -2x$$

$$v_1 = -\cos n\pi x / n\pi$$

$$u'' = -2$$

$$v_2 = -\sin n\pi x / n^2 \pi^2$$

$$v_3 = \cos n\pi x / n^3 \pi^3$$



$$= \left[ (4-x^2) \left( -\frac{\cos n\pi x}{n\pi} \right) - (-2x) \left( -\frac{\sin n\pi x}{n^2\pi^2} \right) + (-2) \left( \frac{\cos n\pi x}{n^3\pi^3} \right) \right]_0^2$$

$$= \left[ \left\{ 0 - \frac{2}{n^3\pi^3} \right\} - \left\{ -\frac{4}{n\pi} - 0 - \frac{2}{n^3\pi^3} \right\} \right] = \frac{4}{n\pi}$$

$$f(x) = 4 - x^2 = \frac{8}{3} - \frac{4}{n^2\pi^2} \left[ \cancel{x} + \frac{4}{n\pi} \right]$$

Q.9

⇒

Find fourier series for  $f(x) = 1-x^2$  in  $(-1,1)$ 

$$f(-x) = 1 - (-x^2) = 1 - x^2 = f(x)$$

Hence  $f(x)$  is even. $(-l, l) \Rightarrow (-1, 1)$  we get  $l=1$ 

$$a_0 = \frac{1}{l} \int_0^l f(x) dx = \int_0^1 (1-x^2) dx$$

$$= \left[ x - \frac{x^3}{3} \right]_0^1 = 1 - \frac{1}{3} = \frac{2}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos(n\pi x) dx$$

$$= 2 \int_0^1 (1-x^2) \cos n\pi x dx$$

$$= 2 \left[ (1-x^2) \left( \frac{\sin n\pi x}{n\pi} \right) - (-2x) \left( -\frac{\cos n\pi x}{n^2\pi^2} \right) + (-2) \left( -\frac{\sin n\pi x}{n^3\pi^3} \right) \right]_0^1$$

$$\therefore u = (1-x^2)$$

$$v = \cos n\pi x$$

$$u' = -2x$$

$$v_1 = \sin n\pi x / n\pi$$

$$u'' = -2$$

$$v_2 = -\cos n\pi x / n^2\pi^2$$

$$u''' = 0$$

$$v_3 = -\sin n\pi x / n^3\pi^3$$

$$a_n = 2 \left[ \frac{-2 \cos n\pi}{n^2\pi^2} \right] = \frac{-4(-1)^n}{n^2\pi^2}$$

$$\therefore b_n = 0$$

$$f(x) = \frac{2}{3} - \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos n\pi x$$

$$= a_0 + \sum \frac{a_n \cos(n\pi x)}{l}$$

Q.10  $\Rightarrow$  Find fourier expansion for  $f(x) = x, -1 \leq x \leq 1$

Here  $l = 1$

$f(x) = x$  which is odd.  $a_0 = 0$  &  $a_n = 0$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

$$= \frac{2}{1} \int_0^1 x \sin n\pi x dx$$

$$= 2 \left[ x \left( -\frac{1}{n\pi} \cos n\pi x \right) - (1) \left( -\frac{1}{n^2\pi^2} \sin n\pi x \right) \right]_0^1$$

$$= 2 \left[ -\frac{1}{n\pi} (-1)^n \right] = -\frac{2}{n\pi} (-1)^n$$

$$f(x) = \frac{-2(-1)^n \sin n\pi x}{n\pi}$$